Getting started in Research and Analytics

Session 6: Matching

Matching

In this session, we'll cover Matching, a balancing measure.

We'll learn that it gives more accurate treatment effects.

Why should we learn this?

To strengthen our studies with treatment and control groups.

To get accurate treatment effects on cost, LOS, SSIs, mortality, etc.



Propensity Score Matching

Background

Issue

Confounding can lead to inaccurate results.

Confounding is when uncontrolled factors have an impact on the predictor variables. Instead of X -> Y

We might have W -> X -> Y. If so, we might incorrectly state the effect of X on Y. There is a limit to how far Regression can balance data.

These are often factors that contribute to whether a patient participates in an intervention group such as age and SES. Or in math tests in schools, Motivation -> Math Intervention -> Math test scores.

Unlike randomization, we may not have a balance in important and unobserved variables in observational research.

Propensity scores give us better balance by making a logistic regression on Treatment group participation and using that probability as a control in our analysis.

Purpose

Propensity scores help to match cases and controls by their risk levels.

Model mis-specification can lead to incorrect estimates of the effect of an intervention.

By not adjusting for risk, we can over/underestimate the true effect of the intervention.

Propensity score approach

Develop a logistic regression model to calculate the propensity score (i.e., the probability of receiving the intervention as a function of the risk factors).

Rank the population from lowest to highest propensity score.

Divide the population into a few even sized groups such as quintiles, with similar propensity scores. More than 5 bins (e.g., deciles) will not provide any more useful information than 5 (Rosenbaum and Rubin, 1984).

Within each propensity score quintile, divide people into the intervention and control groups. Controls are most common in the lower quintiles (where the predicted probability of receiving the treatment is smallest).

Propensity score check

Examine the distribution of risk factors between the intervention and control cases in the subsample to directly observe how their distribution of important risk factors compare.

This illustrates one appealing feature of this approach. The purpose of the sampling is to achieve similar intervention and control groups, and the two groups can be examined to verify how similar their risk factors are by comparing descriptives between the original results and the propensity risk-adjusted model results.

In other words, compare the means of your propensity model predictors. If done correctly, the mean age, heart rate, etc. should be similar for the treatment and control groups.

Matching approaches

Nearest neighbor in which intervention participants are randomly ordered and assigned the match with their nearest score.

Approaches can be improved by setting a “caliper” (maximum acceptable difference in propensity scores).

More sophisticated approaches such as Mahalnobis metric matching either with or without calipers or by using optimization models (see Rosenbaum & Rubin, 1985 and Rosenbaum, 1989). (p.283).

We can do 1 treatment to 1 control case, 1:Many, or Many:1 matching.

Using the propensity score in a regression model

- Fit a new model to predict the effect of the intervention on the combined subsample (and compare with your original sample's results).
- Because the intervention cases and controls in the subsample are similar with respect to baseline factors, raw and risk-adjusted estimates of the treatment effect should be similar.
- Although modeling can reduce the effect of the residual differences between the subsampled groups, its main purpose at this stage is to produce a more accurate estimate (i.e., a narrower confidence intervals) of the treatment effect.

Propensity Score Matching

Example



Study to Understand Prognoses and Preferences for Outcomes and Risks of Treatments (SUPPORT by Frank Harrell)

- Study to examine the outcomes of seriously ill hospitalized patients.
- Data collected from 1991-1994. N= 9,105
- The outcomes are cost and length of stay.
- Adding artificial “program” data to represent an intervention program for matching purposes and treatment effect.

Option 1: Start with creating the propensity score model

R

```
# Get propensity score

propensity_score <- glm(program ~ age + sex + num.co + dzgroup, data = support3, family = binomial(link = "logit"))

# Add scores back to data

support3$ps <- predict(propensity_score, type = "response")
```

Python (see code for libraries)

```
# Rename the column to run in the model

support3 =  support3.rename(columns={'num.co': 'num_co'})

# object for the logistic regression model

propensity_score = smf.glm('program ~ age + sex + num_co + dzgroup', data=support3, family=sm.families.Binomial()).fit()

# Summary of the model results

print(propensity_score.summary())

# Extract propensity scores and add them back to the data

support3['ps'] = propensity_score.predict(support3)
```

Option 2: Match with R/Python packages that make propensity scores and does matching

R

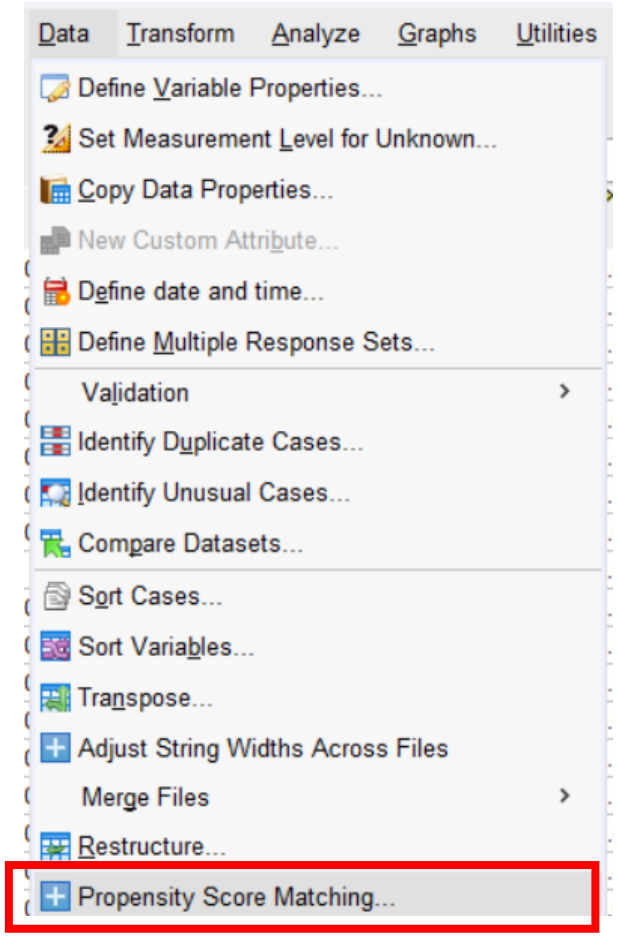
```
matched_data <-
                matchit(program ~ age + sex + num.co + dzgroup, data= support3, method = "nearest", distance = "mahalanobis", replace = TRUE)


# Get a summary, check on balance
summary(matched_data)
```

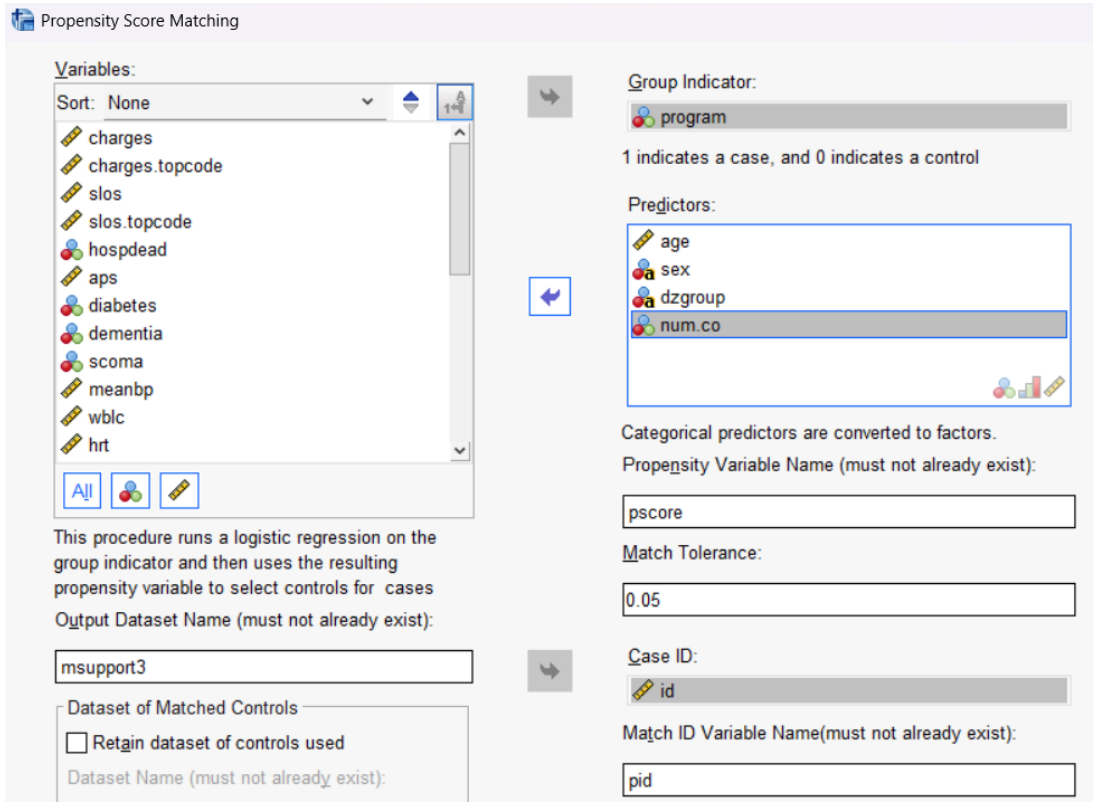
Python

```
# Initialize the PsmPy class
psm = PsmPy(sup3, treatment='program', indx='id', exclude = ['sex', 'dzgroup'])
# Calculate the propensity score
psm.logistic_ps(balance = True)
# Perform K Nearest Neighbor matching
psm.kdtree_matched(matcher='propensity_logit', replacement=False, caliper=None, drop_unmatched=True)
# Create a matched dataset
sup3_match = psm.df_matched
# Remove cases that were not matched
msupport3 = support3[support3['id'].isin(sup3_match['id'])]
```

Propensity score matching in SPSS



It took 15 minutes to run this step below



Case Control Matching Statistics[a]

Match Type	Count
Demander Cases	4105
Supplier Cases	5000
Matches	4088
Unmatched	17
Sampling	without replacement
Log file	none
Demander Queue Limit	4106

a. Match variables: pscore

Check the Summary of Balance for Matched Data from R

Excellent: Almost identical means for treatment and control groups, no statistically significant differences.

	Means Treated	Means Control
age	62.6645	62.6714
sexfemale	0.4219	0.4219
sexmale	0.5781	0.5781
num.co	2.0156	2.0061
dzgroupARF/MOSF w/Sepsis	0.3496	0.3496
dzgroupCHF	0.1459	0.1459
dzgroupCirrhosis	0.0629	0.0629
dzgroupColon Cancer	0.0443	0.0443
dzgroupComa	0.0787	0.0787
dzgroupCOPD	0.1303	0.1303
dzgroupLung Cancer	0.1072	0.1072
dzgroupMOSF w/Malig	0.0811	0.0811

This looks good, we're now ready to analyze the data.

Comparison of the propensity score and 3 weight variables for patients by intervention groups

- Higher weight values have more influence in the regressions.
- Generally, the intervention group has more weight when pscore < 0.50.
- And the control group has more weight when pscore > 0.50.

	Group	pscore	ipw	nipw	att
1	Control	0.364	1.57	0.00017	0.572
2	Intervention	0.363	2.76	0.0003	1
3	Control	0.464	1.86	0.00021	0.864
4	Intervention	0.462	2.17	0.00024	1
5	Control	0.535	2.15	0.00024	1.15
6	Intervention	0.528	1.89	0.00021	1

Summary of R results

	Savings	p-value	N
None	-34,927	p<0.001	9105
Matched	-33,674	p<0.001	6471
Propensity	-31,731	p<0.001	9105
IPW	-31,430	p<0.001	9105
NIPW	-31,430	p<0.001	9105
ATT	-31,874	p<0.001	9105

- Using matching or weights provides more accurate estimates.
- This conservatively helps reduce estimated savings by $3,500 or $35 million per 10,000 patients.
- Using the propensity score and IPW weights uses the full sample.

ham package code for a regression using the propensity score or inverse probability weights

Covariate Adjustment with the propensity score

```
mps <- assess(charges ~ program ,
data=support3,

intervention = "program",
regression="ols",
propensity=c("age", "sex",
"num.co", "dzgroup"))

summary(mps$model)
```

Inverse Probability Weights model

```
mipw <- assess(charges ~ program,
data=support3,

intervention = "program",
regression="ols",
propensity=c("age", "sex",
"num.co", "dzgroup"),
weights="ipw")

summary(mipw$model)
```

No need to create the propensity score model and analyze separately, ham does it all in 1 step.

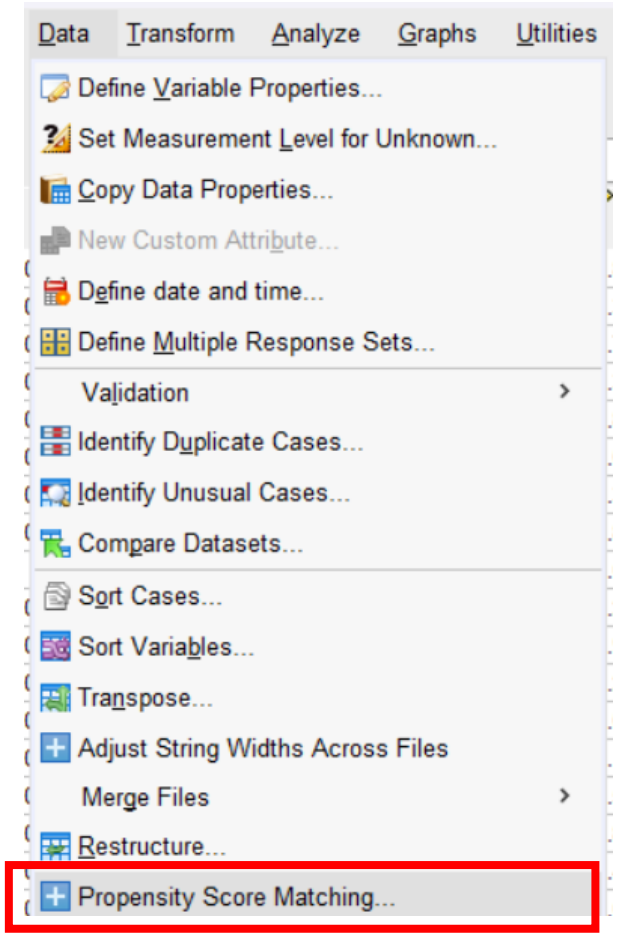
Summary

- Propensity score matching allows us improve how we control for confounding in observational studies.
- This helps us get more accurate treatment effects.
- The same treatment effects can be calculated in matching (e.g., t-test) or as a regression coefficient.
- There are many options in matching.
- Using the covariate adjustment with the propensity score (or inverse probability weights) offers an easy approach that makes use of the full sample while providing conservative estimates that may hold up well over time (Harrell, 2015).
- See “An Introduction to Propensity Score Methods for Reducing the Effects of Confounding in Observational Studies” (Austin, 2011) article in the StatsWithSteve -> Articles & Books folder for more on matching.

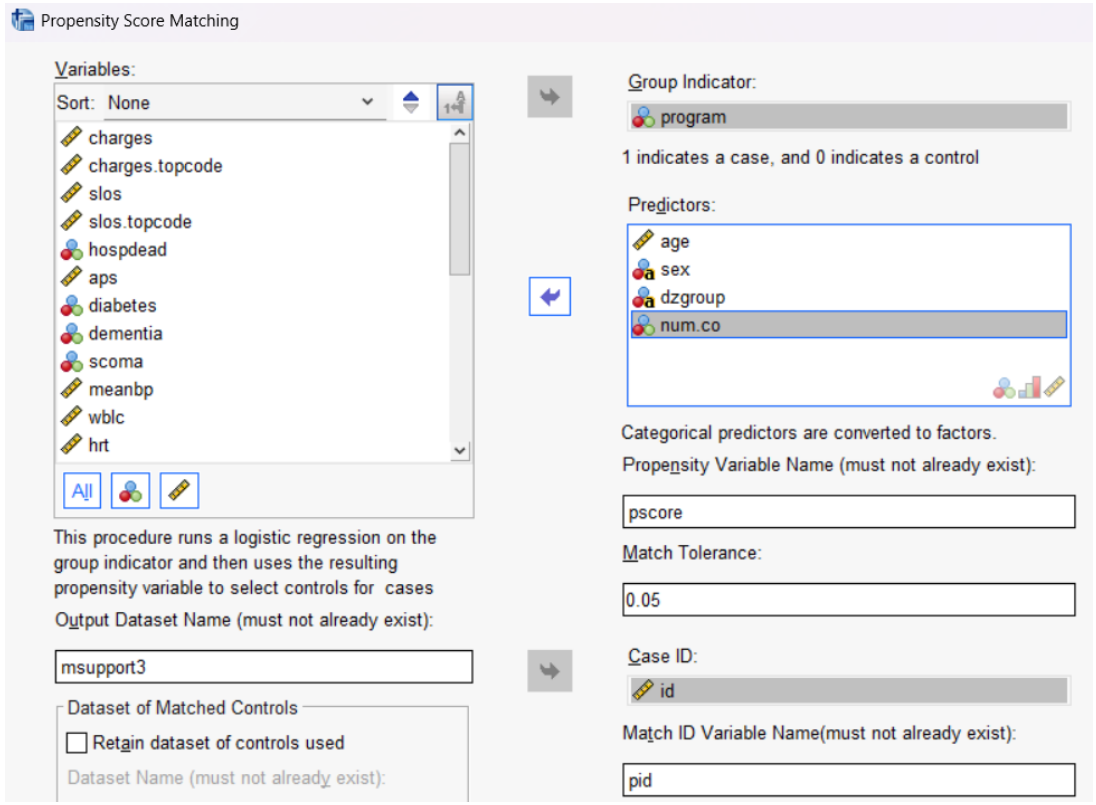
Appendix A

SPSS example for matching, t-test, and a linear regression

Propensity score matching



It took 15 minutes to run this step below



Case Control Matching Statistics[a]

Match Type	Count
Demander Cases	4105
Supplier Cases	5000
Matches	4088
Unmatched	17
Sampling	without replacement
Log file	none
Demander Queue Limit	4106

a. Match variables: pscore

Results from an independent sample t-test and linear regression with propensity score (can do a paired-sample with both pairs on the same row)

N = 8,064

Independent Samples Test

		Levene's Test for Equality of Variances		t-test for Equality of Means							
						Significance				95% Confidence Interval of the Difference	
		F	Sig.	t	df	One-Sided p	Two-Sided p	Mean Difference	Std. Error Difference	Lower	Upper
charges	Equal variances assumed	470.387	<.001	-16.283	8063	<.001	<.001	-37570.41550	2307.3090319	-42093.33705	-33047.49395
	Equal variances not assumed			-16.262	5743.220	<.001	<.001	-37570.41550	2310.3860365	-42099.64344	-33041.18756

Coefficients[a]

Model		Unstandardized Coefficients		Standardized Coefficients		
		B	Std. Error	Beta	t	Sig.
1	(Constant)	137712.795	6474.446		21.270	<.001
	program	-34595.193	2316.221	-.164	-14.936	<.001
	Predicted probability	-133710.325	14138.032	-.104	-9.457	<.001

a. Dependent Variable: charges

Propensity score